

# Probability

A review, Weak Law of Large Numbers and Central Limit Theorem

---

Noara Razzak

9 May, 2020

**What is Probability?** Probability measures **how likely** something is to happen, from:

- **0** (never happens)
- to **1** (always happens) (or 0% to 100%)

Rolling a fair die:

- $P(\text{Even}) = \frac{3}{6} = 0.5$
- $P(\text{Prime}) = \frac{3}{6} = 0.5$

## Basic Probability Formula

$$P(\text{Event}) = \frac{\text{Number of ways it can happen}}{\text{Total possible outcomes}}$$

## Important rules

### Certain Event (100%)

$$P(\text{Sun rises}) = 1$$

### Impossible Event (0%)

$$P(\text{Rolling a 7 on standard die}) = 0$$

### Complement Rule ("Not")

$$P(\text{Not } A) = 1 - P(A)$$

### Conditional Probability ("If...Then")

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)}$$

## Rules of probability

| Rule               | Formula                                   |
|--------------------|-------------------------------------------|
| Complement         | $P(A^c) = 1 - P(A)$                       |
| Union              | $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ |
| Mutually Exclusive | $P(A \cap B) = 0$                         |
| Independent        | $P(A \cap B) = P(A)P(B)$                  |

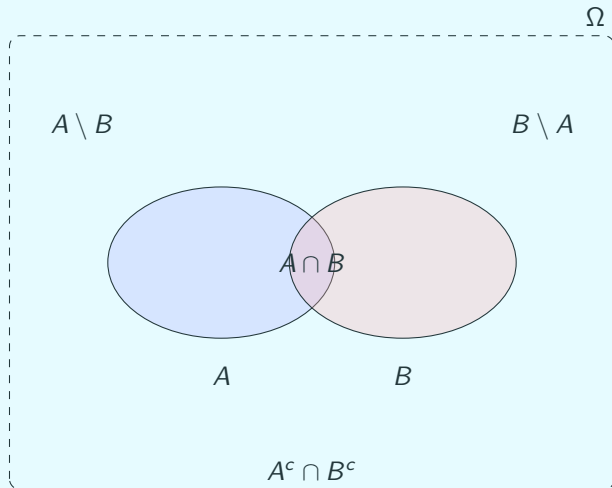
## Bayes' Theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

## Bayes' Theorem originates from Conditional Probability

The probability of  $A$  *given*  $B$ :

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$



## Real-world uses

- Medical diagnosis
- Spam filtering
- Risk assessment
- Machine learning (Naive Bayes)

## Key Insight

- Conditional probability formalizes how we should rationally update beliefs given evidence.

## Example 1

A family has two children. You know that at least one of them is a boy. What is the probability that both children are boys?

## Example 1 - solution

**Step 1: Sample Space** The possible gender combinations for two children (older child first) are:

$$S = \{BB, BG, GB, GG\}$$

**Step 2: Apply Condition** We know *at least one* is a boy, so we eliminate GG:

$$S' = \{BB, BG, GB\}$$

**Step 3: Count Favorable Outcomes** Only BB has two boys:

$$\text{Favorable} = 1 \quad (BB)$$

$$\text{Possible} = 3 \quad (BB, BG, GB)$$

**Step 4: Calculate Probability**

$$P(\text{Both boys}) = \frac{\text{Favorable}}{\text{Possible}} = \frac{1}{3}$$

## Example 2

A bag contains 4 red marbles, 5 blue marbles, and 6 green marbles. If 3 marbles are randomly selected without replacement, what is the probability that all three are blue?

- $\frac{1}{91}$
- $\frac{2}{91}$
- $\frac{5}{182}$
- $\frac{10}{273}$
- $\frac{25}{546}$

**Example 2 - solution Step 1: Determine Total Marbles** First, find the total number of marbles:

$$4 \text{ red} + 5 \text{ blue} + 6 \text{ green} = 15 \text{ marbles}$$

**Step 2: Calculate Total Possible Outcomes** Number of ways to choose any 3 marbles from 15:

$$\text{Total combinations} = \binom{15}{3} = \frac{15!}{3!(15-3)!} = 455$$

**Step 3: Calculate Favorable Outcomes** Number of ways to choose 3 blue marbles from 5 available:

$$\text{Favorable combinations} = \binom{5}{3} = \frac{5!}{3!(5-3)!} = 10$$

**Step 4: Compute Probability**

$$P(3 \text{ blue}) = \frac{\text{Favorable}}{\text{Total}} = \frac{10}{455} = \frac{2}{91}$$

– (Correct Answer)

## Weak Law of Large Numbers (WLLN)

The **Weak Law of Large Numbers (WLLN)** is a fundamental theorem in probability theory that describes how the sample average of a sequence of independent and identically distributed (i.i.d.) random variables converges in probability to the expected value (i.e. population mean) as the sample size increases.

## Statement of the Theorem

Let  $X_1, X_2, \dots, X_n$  be a sequence of i.i.d. random variables with finite mean  $\mu = \mathbb{E}[X_i]$  and finite variance  $\sigma^2 = \text{Var}(X_i)$ . Define the sample mean as:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

The WLLN states that for any  $\epsilon > 0$ ,

$$\lim_{n \rightarrow \infty} \mathbb{P}(|\bar{X}_n - \mu| \geq \epsilon) = 0.$$

In other words,  $\bar{X}_n$  **converges in probability** to  $\mu$ , denoted as:

$$\bar{X}_n \xrightarrow{P} \mu.$$

## Example

Suppose  $X_1, X_2, \dots$  are i.i.d. Bernoulli random variables with success probability  $p = 0.5$ . Then:

$$\mu = \mathbb{E}[X_i] = p = 0.5, \quad \sigma^2 = p(1 - p) = 0.25.$$

The WLLN guarantees that for large  $n$ , the sample mean  $\bar{X}_n$  will be very close to 0.5 with high probability. For instance, if  $n = 10,000$  and  $\epsilon = 0.01$ :

$$\mathbb{P}(|\bar{X}_n - 0.5| \geq 0.01) \leq \frac{0.25}{10,000 \times 0.01^2} = 0.25.$$

As  $n$  increases, this probability decreases further.

## Interpretation

- The WLLN justifies the intuitive idea that averaging many independent observations reduces noise.
- It is "weak" because it guarantees convergence in probability (not almost surely, which is the Strong Law of Large Numbers).
- Applications include:
  - Polling and surveys (sample proportions converge to population proportions).
  - Monte Carlo methods (approximating integrals via random sampling).

## Comparison with Strong Law

| Weak Law (WLLN)                 | Strong Law (SLLN)                  |
|---------------------------------|------------------------------------|
| $\bar{X}_n \xrightarrow{P} \mu$ | $\bar{X}_n \xrightarrow{a.s.} \mu$ |
| Convergence in probability      | Almost sure convergence            |
| Weaker guarantee                | Stronger guarantee                 |

**Table 1:** WLLN vs. SLLN

## Central Limit Theorem

Two fundamental theorems in probability theory—the **Weak Law of Large Numbers (WLLN)** and the **Central Limit Theorem (CLT)**—describe the behavior of sample averages. While the WLLN establishes convergence, the CLT refines this by specifying the *distribution* of the sample mean around the population mean. This document explores their connection.

## Central Limit Theorem (CLT)

Under the same conditions (i.i.d. with  $\mu, \sigma^2 < \infty$ ), the CLT states:

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2),$$

where  $\xrightarrow{d}$  denotes convergence in distribution, and  $\mathcal{N}(0, \sigma^2)$  is a normal distribution with mean 0 and variance  $\sigma^2$ .

**Interpretation** The CLT provides a *distributional approximation* for the sample mean:

$$\bar{X}_n \approx \mu + \frac{\sigma}{\sqrt{n}}Z, \quad Z \sim \mathcal{N}(0, 1).$$

This refines the WLLN by quantifying the rate of convergence and the shape of fluctuations.

## Mathematical Link

The WLLN can be derived from the CLT:

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \epsilon) = \mathbb{P}(|\sqrt{n}(\bar{X}_n - \mu)| \geq \epsilon\sqrt{n}) \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

because  $\epsilon\sqrt{n} \rightarrow \infty$ , and the CLT ensures  $\sqrt{n}(\bar{X}_n - \mu)$  is stochastically bounded

## Joint Implications in Statistical Inference -Confidence Intervals

- **WLLN**: Justifies using  $\bar{X}_n$  as an estimator for  $\mu$ .
- **CLT**: Provides the approximate distribution for constructing confidence intervals:

$$\mu \in \left[ \bar{X}_n \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right].$$

## Connection Between WLLN and CLT

- **WLLN**: Ensures  $\bar{X}_n$  converges to  $\mu$  (consistency).
- **CLT**: Describes the *scaled deviations*  $\sqrt{n}(\bar{X}_n - \mu)$  (asymptotic normality).

## Intuition

- The WLLN shows that  $\bar{X}_n - \mu \rightarrow 0$ , but the CLT reveals that the **rate** is  $\mathcal{O}(n^{-1/2})$ :

$$|\bar{X}_n - \mu| \approx \frac{\sigma}{\sqrt{n}}|Z|.$$

- The CLT implies the WLLN (since convergence in distribution to a constant implies convergence in probability), but not vice versa.

## Comparison Summary

| Feature          | WLLN                              | CLT                                                       |
|------------------|-----------------------------------|-----------------------------------------------------------|
| Convergence Type | Probability ( $\xrightarrow{P}$ ) | Distribution ( $\xrightarrow{d}$ )                        |
| Rate             | $\bar{X}_n - \mu \rightarrow 0$   | $\sqrt{n}(\bar{X}_n - \mu) \sim \mathcal{N}(0, \sigma^2)$ |
| Key Use          | Consistency of estimators         | Confidence intervals,<br>hypothesis testing               |

**Table 2:** WLLN vs. CLT

**Conclusion** The WLLN and CLT are deeply connected: the CLT *extends* the WLLN by providing a distributional limit for the sample mean. Together, they underpin modern statistical inference, ensuring both consistency and asymptotic normality of estimators.

**Next class we will cover confidence intervals.**